



University of
Nottingham

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COMP4139

Machine Learning (MLE)

Deep Convolutional Neural Networks

- Part 1

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- What is unsupervised learning?
- How do we measure similarity between vectors?
- K-Means Clustering?
- Agglomerative Hierarchical Clustering?
- Density-based Spatial Clustering of Applications with Noise (DBSCAN)?
- Expectation Maximisation?
- Silhouette Co-efficient?
- Dunn's Index?



Learning Objectives

- What is a Convolutional Neural Network?
- What is a convolution?
- ~~What is max pooling?~~
- How do we handle color images with neural networks?
- What is vanishing gradient?
- Assignment 2 discussion

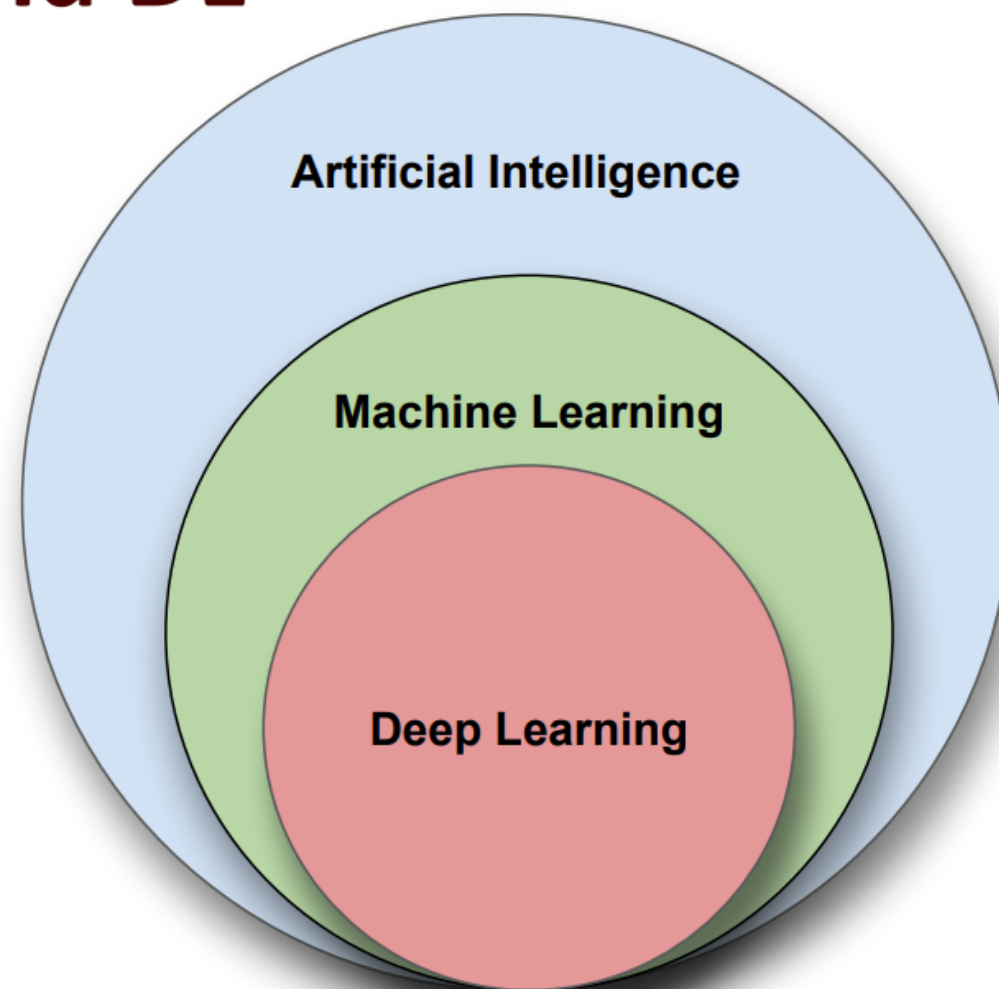


What is deep learning?

- Based on the principle of Artificial Neural Networks but more (deeper) layers, more neurons, more data.
- Popular Methods:
 - Convolutional Neural Networks (object recognition, regression, image segmentation, mainly based on visual content: images, video, etc.)
 - Reinforcement Learning Networks (outputs actions, e.g. AI game, robots, etc.)
 - Generative Neural Networks (dimensionality reduction, manifold learning, content synthesis, etc.)
 - Recurrent Neural Networks (sequence to sequence learning, e.g. language translation, etc.)
 - Graph Neural Networks (handles unstructured data, e.g. social networks, etc.)

Relationship of AI, ML and DL

- **Artificial Intelligence (AI)** is anything about man-made intelligence exhibited by machines.
- **Machine Learning (ML)** is an approach to achieve **AI**.
- **Deep Learning (DL)** is one technique to implement **ML**.





Why deep learning?

Problems with traditional ML:

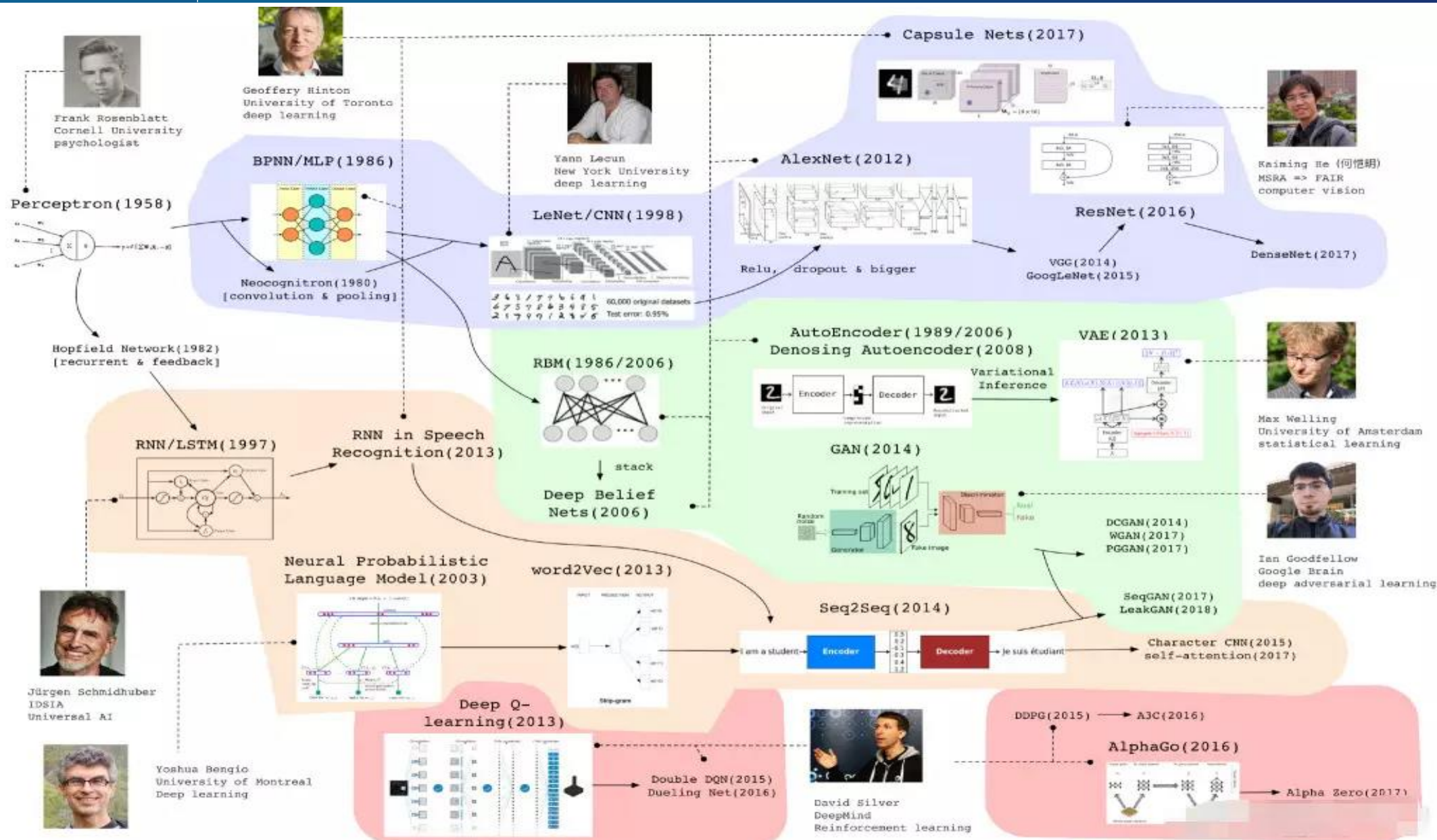
1. Handling high dimensional data
2. Manual feature engineering

How DL helps:

1. Automatic feature detection
2. Computationally expensive, but handles high dimensional data

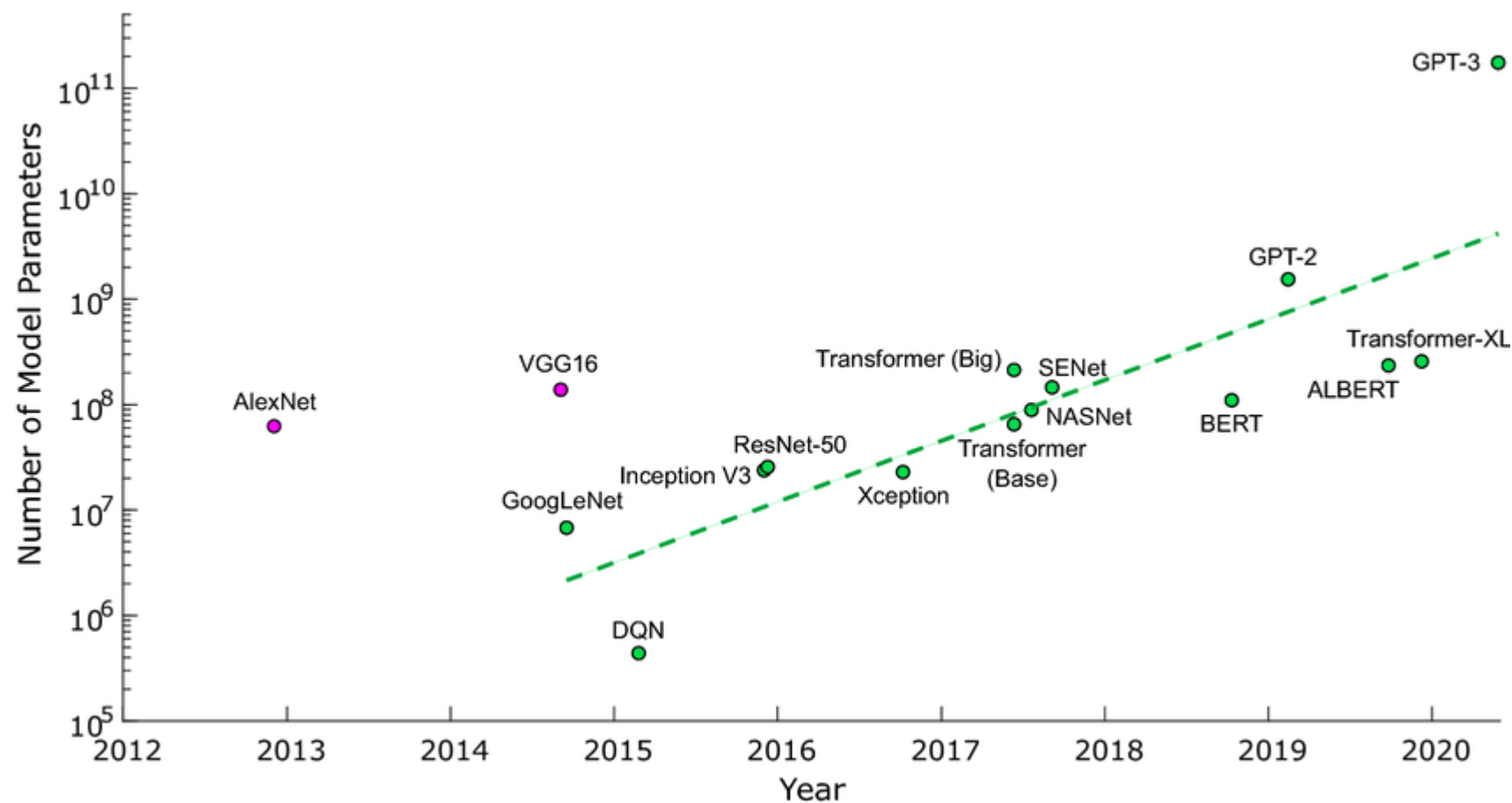


Deep Learning Family

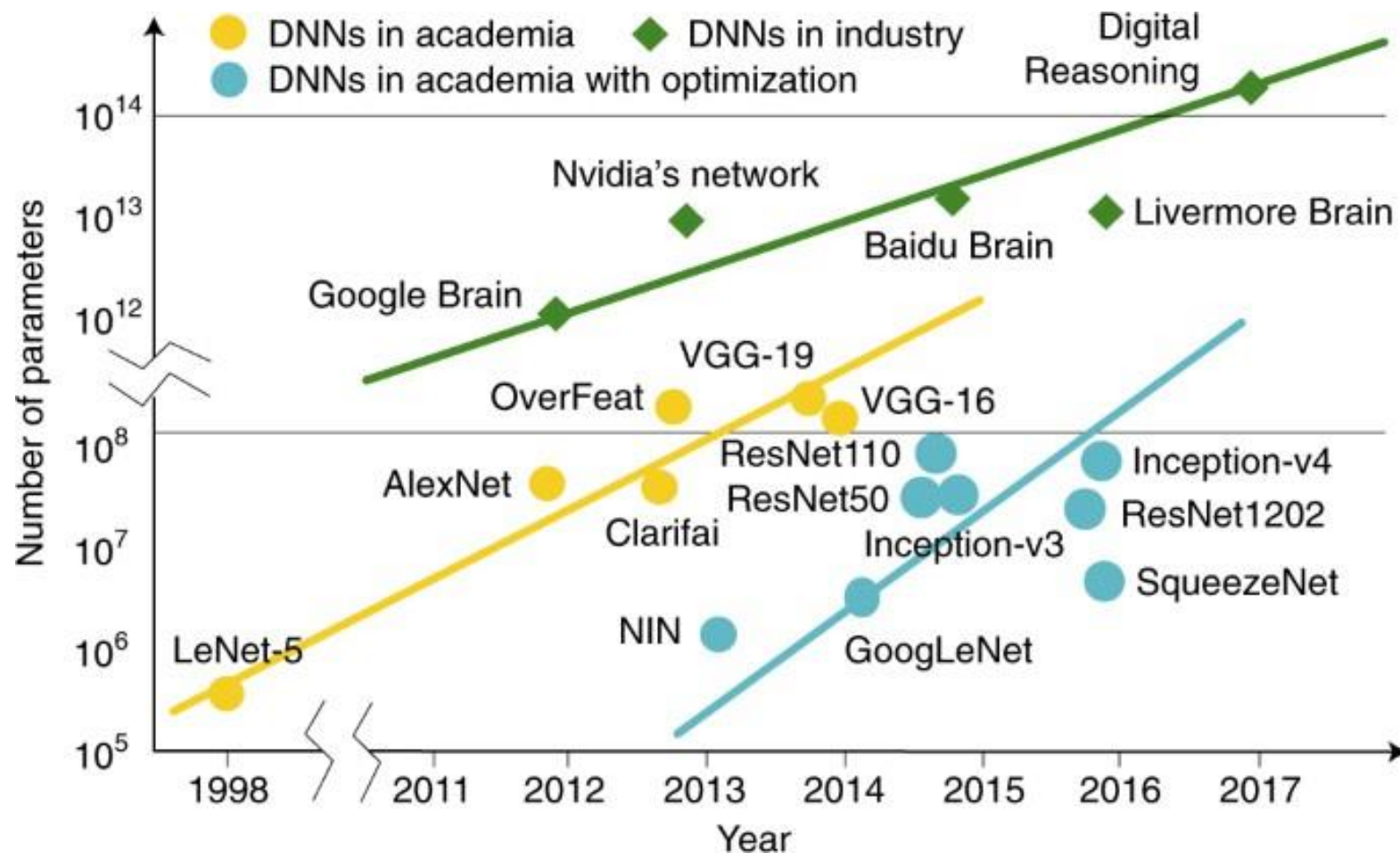




Deep Learning over time

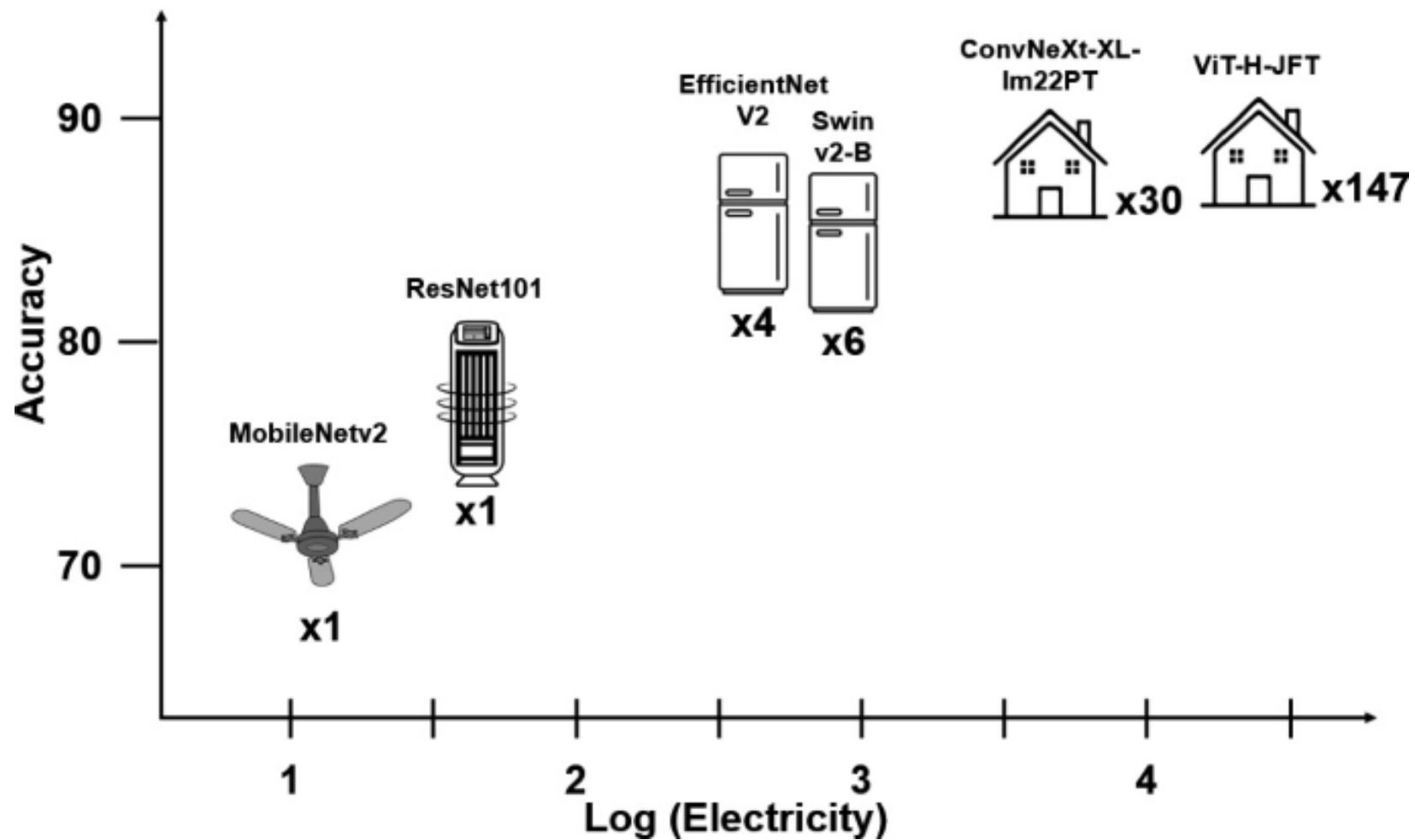


Deep Learning over time

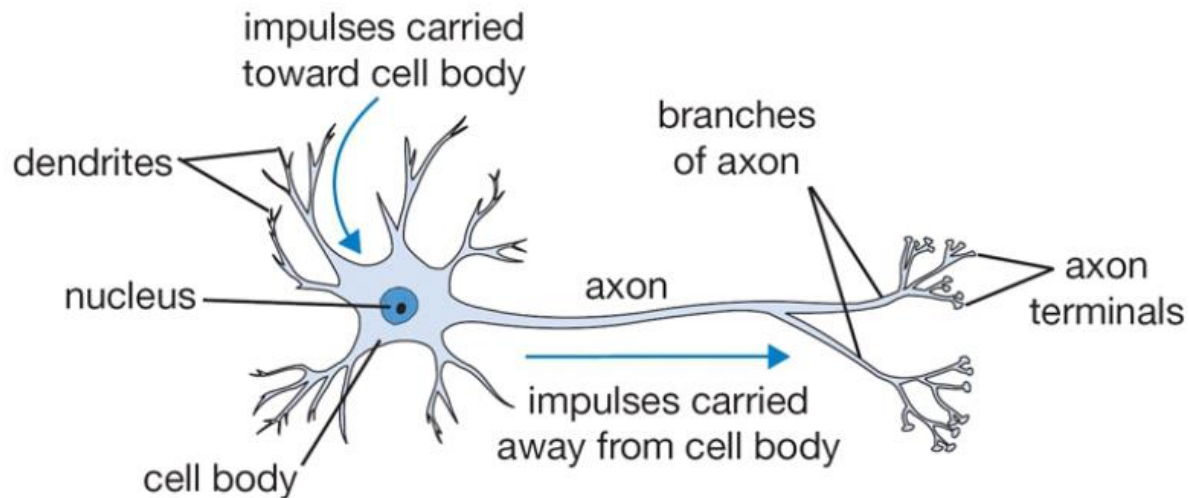




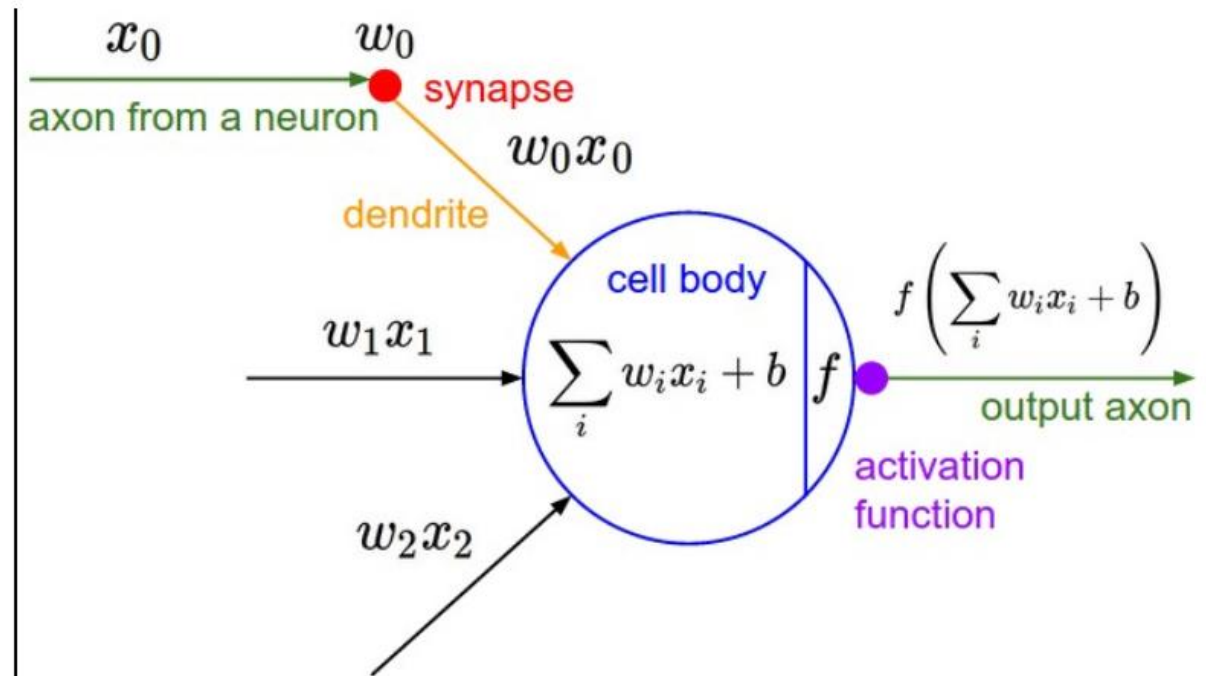
Deep Learning over time



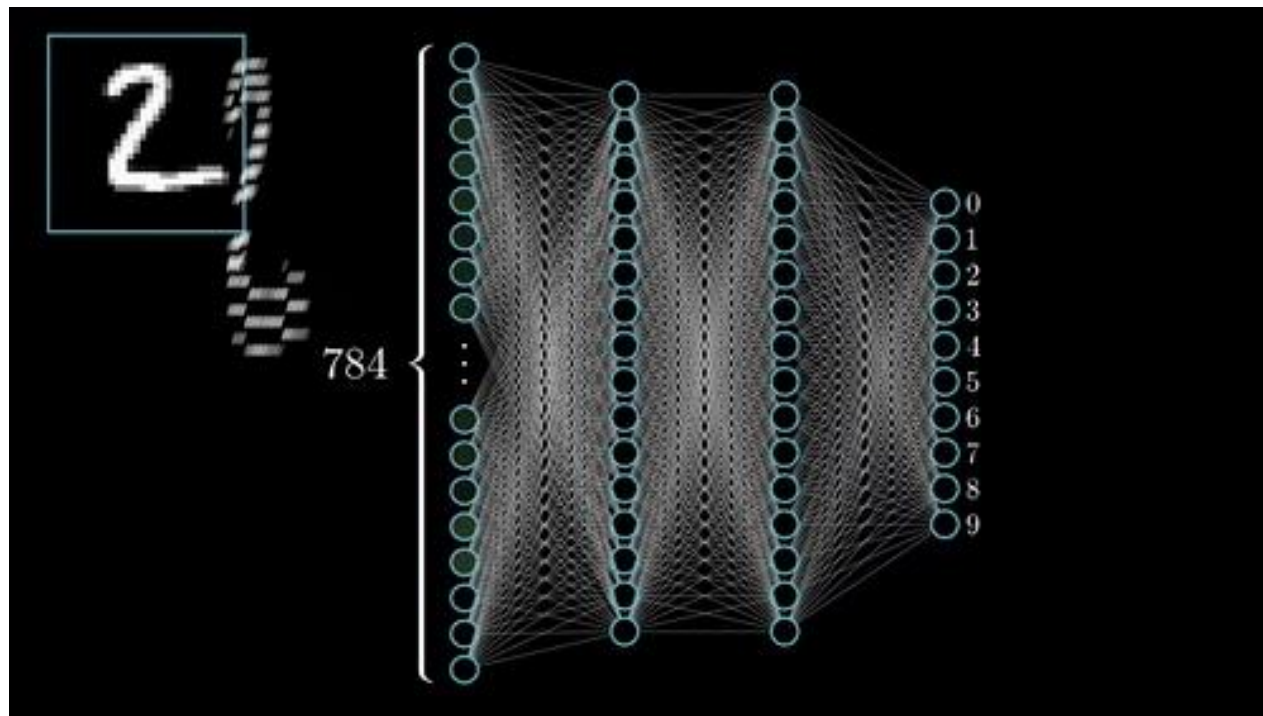
- Biological Motivation
 - ~86 billion neurons in human's nervous system
 - Connected with 10^{14} - 10^{15} synapse



Biological illustration



Mathematical model



Input layer: 784 units (pixels)

Hidden layer 1: 16 units

Hidden layer 2: 16 units

Output layer: 10 units

Ignoring biases.

Trainable parameters:

$$784 \times 16 = 12,544$$

+

$$16 \times 16 = 256$$

+

$$16 \times 10 = 160$$

$$= 12,960$$



64x64

→ Cat?

Input layer: 64x64x3 units (pixels)

Hidden layer 1: 16 units

Hidden layer 2: 16 units

Output layer: 10 units

Ignoring biases.

Trainable parameters:

$$12,288 \times 16 = 196,608$$

+

$$16 \times 16 = 256$$

+

$$16 \times 10 = 160$$

$$= 197,024$$



64x64

→ Cat?

Input layer: 64x64x3 units (pixels)

Hidden layer 1: 16 units

Hidden layer 2: 16 units

Output layer: 10 units

Ignoring biases.

Trainable parameters:

$$12,288 \times 16 = 196,608$$

+

$$16 \times 16 = 256$$

+

$$16 \times 10 = 160$$

$$= 197,024$$

1000x1000
image?





64x64

→ Cat?

Input layer: 1000x1000x3 (pixels)

Hidden layer 1: 16 units

Hidden layer 2: 16 units

Output layer: 10 units

Ignoring biases.

Trainable parameters:

$$3 \times 10^6 \times 16 = 48 \times 10^6$$

+

$$16 \times 16 = 256$$

+

$$16 \times 10 = 160$$

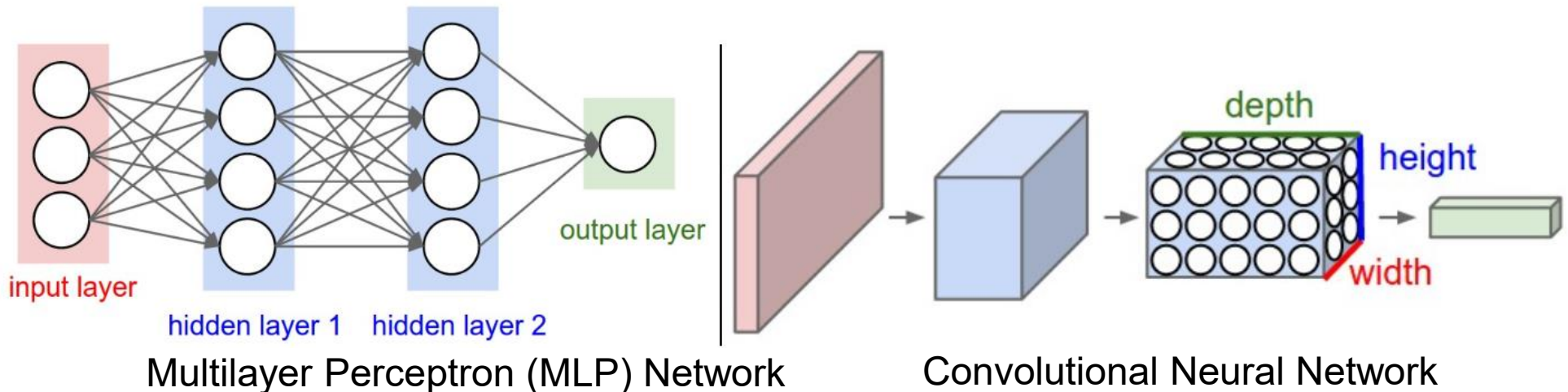
$$\sim 48 \times 10^6$$

1000x1000
image?



Overview of Convolutional Neural Network (CNN)

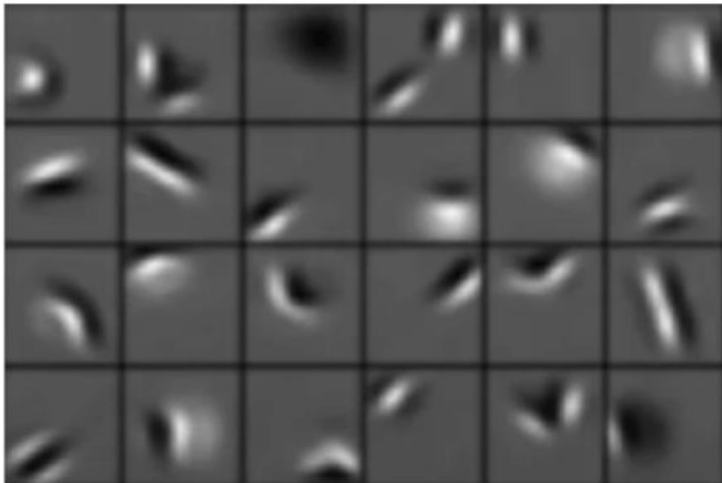
- Multilayer Perceptron Network
 - ✓ Wide application scenario.
 - x Neurons are fully connected- can't scale well to input data with large number of features (e.g. pixels in an images).
- Convolutional Neural Network
 - ✓ Neuron is arranged in "3D", each neuron is only connected by a small region of previous layer
 - ✓ Features are learned in a hierarchical structure: from low level features to high level features.





Low-level to high-level feature visualization

Low-level features



Edges, dark spots

Mid-level features



Eyes, ears, nose

High-level features



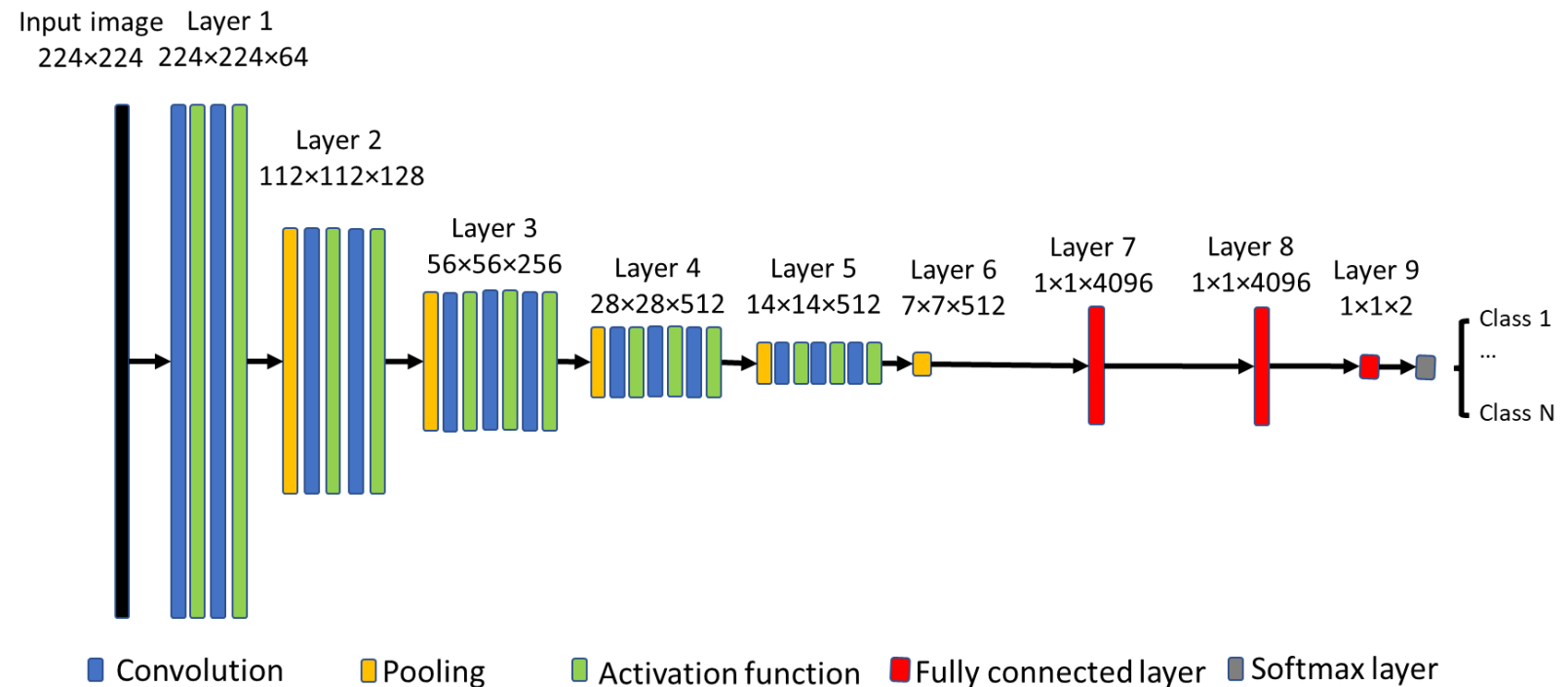
Facial structure



Basic Components of Convolutional Neural Network

- Convolution operator
- Pooling
- Activation function
- Fully connected layer
- Loss functions
- Optimisation methods

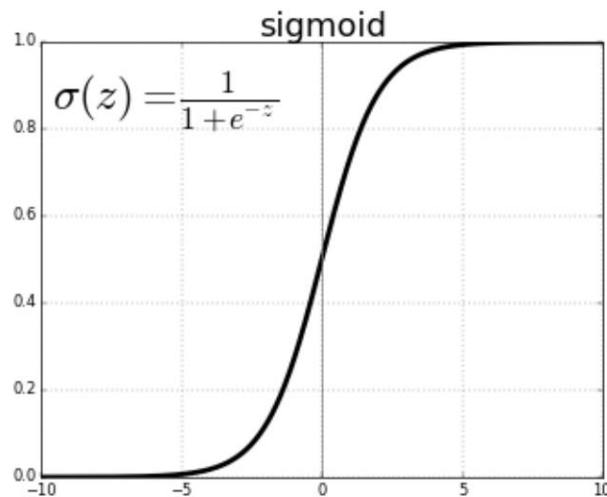
Example Deep CNN for classification



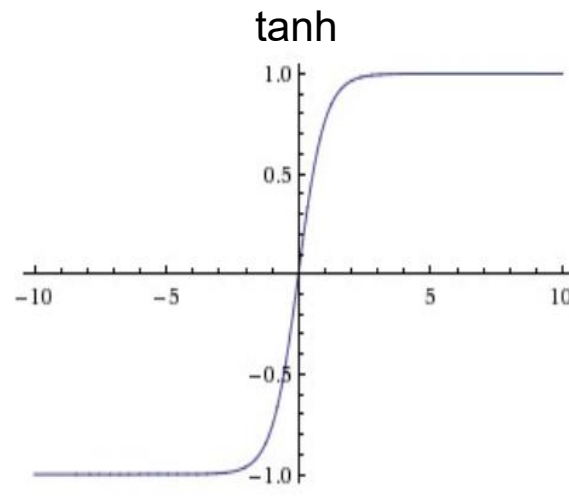


Activation Functions

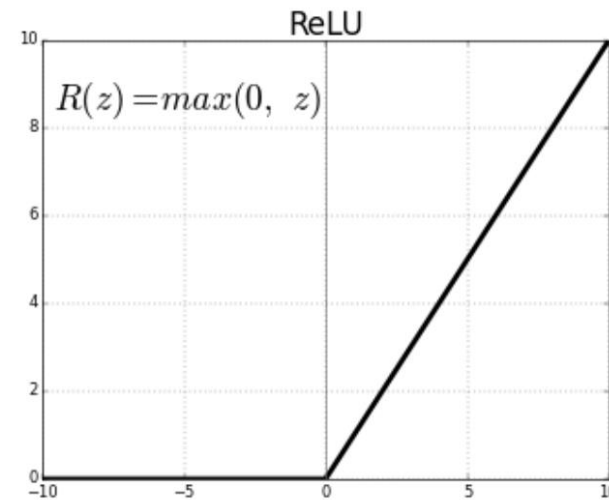
Add nonlinearity to the network, as linear function is useless to learn complicated relationships



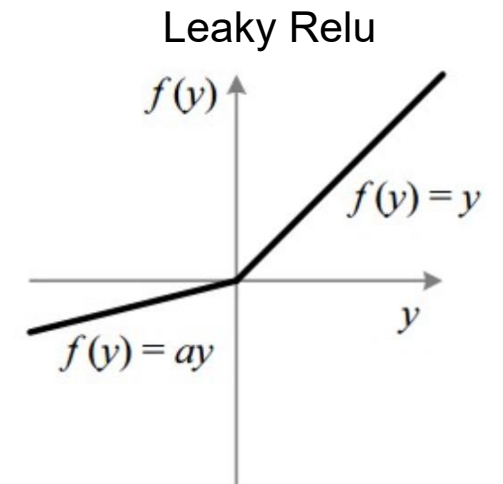
- ✓ Transfer the input range to $[0, 1]$
- x Saturate and kill gradients: small gradient at region of 0 and 1



- ✓ Zero centred range: $[-1, 1]$
- x Saturate and kill gradients: small gradient at region of -1 and 1



- ✓ Non-saturate; converge faster
- ✓ Simple to calculate
- x Some neuron can be dead with negative input (better with smaller learning rate).

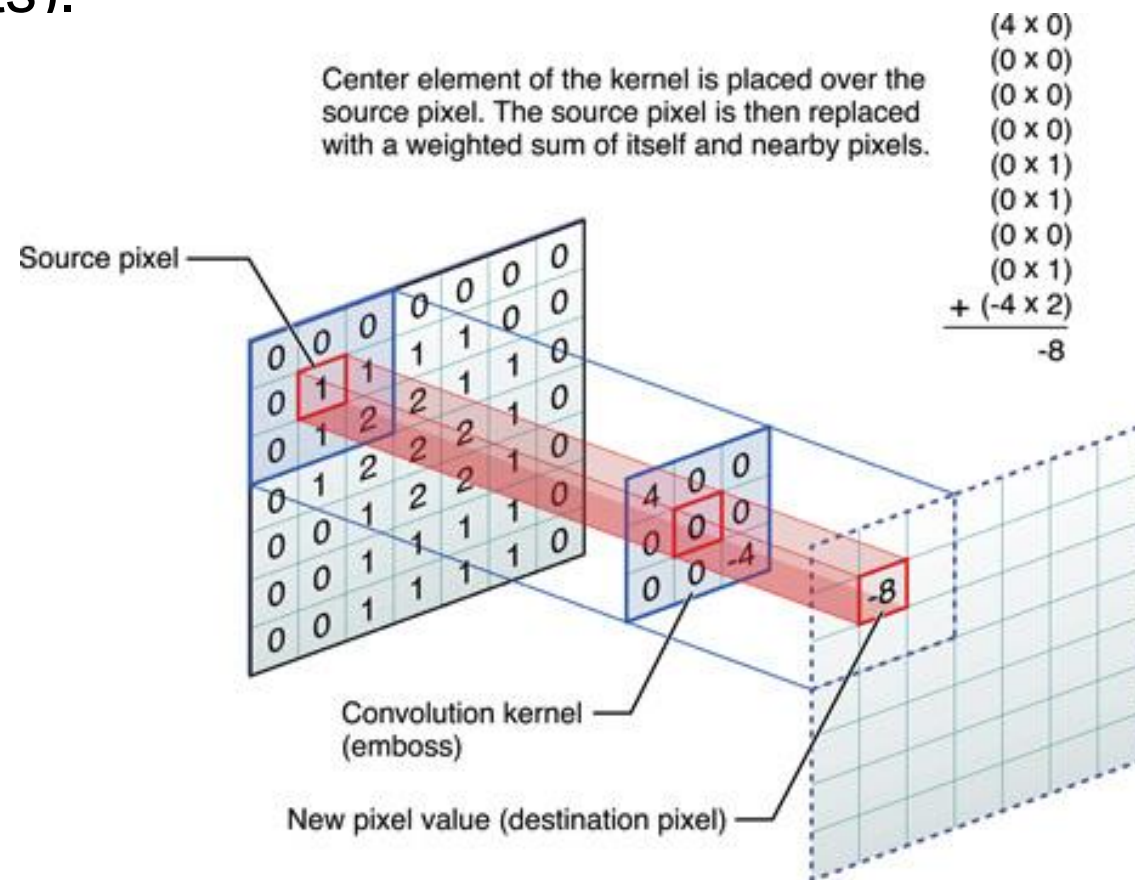
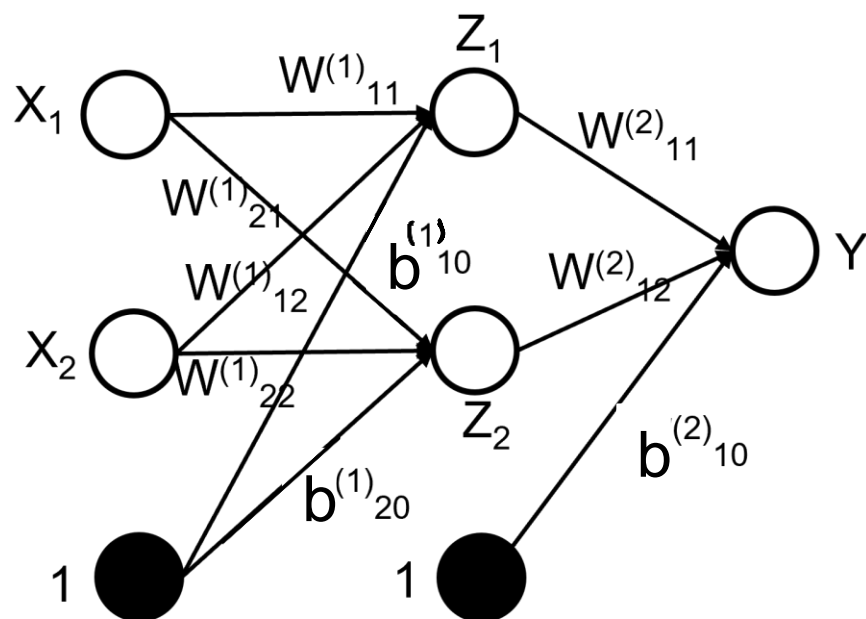


- ✓ Overcome the 'dying neuron'
- x Performance not consistent

Known as gradient vanishing

Convolution Operator

- Convolution is the sum of multiplications of the input data (feature) and a filter (weights).
- It is vector dot product.
- Similar to MLP but in a local region





Edge detection as an example for convolution

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6

*

1	0	-1
1	0	-1
1	0	-1

3x3
filter

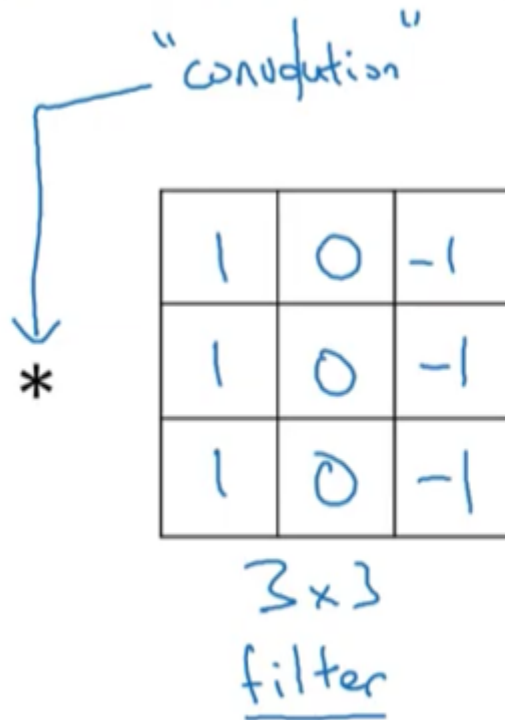


Edge detection as an example for convolution

$$3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1$$

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6



=

4x4



Edge detection as an example for convolution

$$3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1$$

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6

"convolution"

*

1	0	-1
1	0	-1
1	0	-1

3x3
filter

=

-5			

4x4

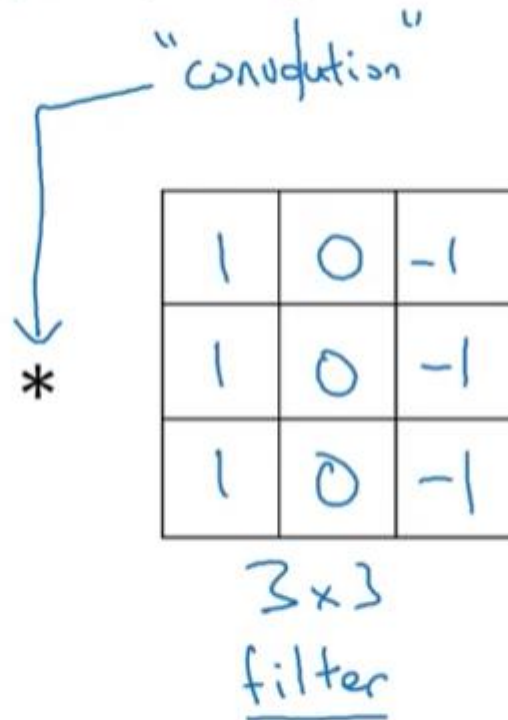


Edge detection as an example for convolution

$$3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 3 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1 = -5$$

3	<u>0</u> ¹	<u>1</u> ⁰	<u>2</u> ⁻¹	7	4
1	<u>5</u> ¹	<u>8</u> ⁰	<u>9</u> ⁻¹	3	1
2	<u>7</u> ¹	<u>2</u> ⁰	<u>5</u> ⁻¹	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6



=

<u>-5</u>	<u>-4</u>		

4x4



Edge detection as an example for convolution

$$3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1 = -5$$

3	<u>0</u>	<u>1</u>	<u>2</u>	<u>7</u>	<u>4</u>
1	<u>5</u>	<u>8</u>	<u>9</u>	<u>3</u>	<u>1</u>
2	<u>7</u>	<u>2</u>	<u>5</u>	<u>1</u>	<u>3</u>
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6

"convolution"

*

1	0	-1
1	0	-1
1	0	-1

3x3
filter

=

<u>-5</u>	<u>-4</u>	0	<u>8</u>
<u>-10</u>	-2	2	3
0	-2	-4	-7
-3	-2	-3	<u>-16</u>

4x4

$$n - f + 1 \times n - f + 1$$
$$6 - 3 + 1 = 4$$



Edge detection as an example for convolution

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0

6x6



*

1	0	-1
1	0	-1
1	0	-1

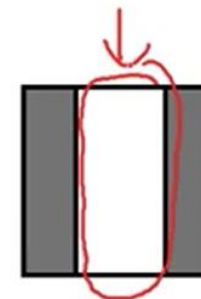
3x3

=

0	30	30	0
0	30	30	0
0	30	30	0
0	<u>30</u>	30	0

4x4

*



Problems with the convolution operator

1. Output shrinking. Can eventually become too small for further processing.
This would be a problem in deep models.
2. Edge pixels do not get much coverage.

Solution:
Padding!

0	0	0	0	0	0	0	0
0	3	3	4	4	7	0	0
0	9	7	6	5	8	2	0
0	6	5	5	6	9	2	0
0	7	1	3	2	7	8	0
0	0	3	7	1	8	3	0
0	4	0	4	3	2	2	0
0	0	0	0	0	0	0	0

$6 \times 6 \rightarrow 8 \times 8$

*

1	0	-1
1	0	-1
1	0	-1

3×3

=

-10	-13	1			
-9	3	0			

6×6

$n+2p-f+1 \times n+2p-f+1$



Types of padding

→ no padding

“Valid”: $n \times n$ $\ast$ $f \times f$ $\rightarrow \frac{n-f+1}{1} \times \frac{n-f+1}{1}$
 6×6 $\ast$ 3×3 $\rightarrow 4 \times 4$

“Same”: Pad so that output size is the same as the input size.

$$n + 2p - f + 1 \times n + 2p - f + 1$$
$$\cancel{n + 2p - f + 1} = \cancel{n} \Rightarrow \underline{p} = \frac{f-1}{2}$$



Stride

2 ³	3 ⁴	7 ⁴	4	6	2	9
6 ¹	6 ⁰	9 ²	8	7	4	3
3 ⁻¹	4 ⁰	8 ³	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

3x3

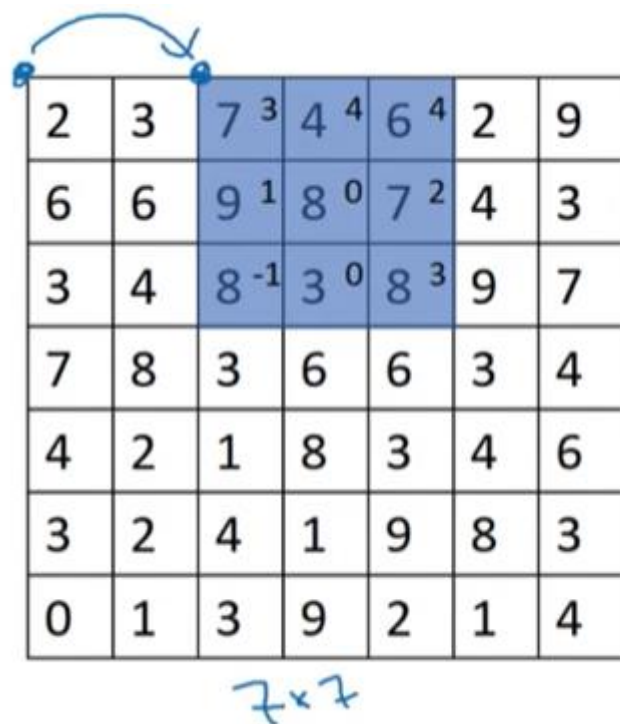
Stride = 2

=

q _i		



Stride



2	3	7 ³	4 ⁴	6 ⁴	2	9
6	6	9 ¹	8 ⁰	7 ²	4	3
3	4	8 ⁻¹	3 ⁰	8 ³	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

3x3

Stride = 2

=

91	100	



Stride

2	3	7	4	6	2	9
6	6	9	8	7	4	3
3 ³	4 ⁴	8 ⁴	3	8	9	7
7 ¹	8 ⁰	3 ²	6	6	3	4
4 ⁻¹	2 ⁰	1 ³	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

3x3

Stride = 2

=

91	100	83
69		



Stride

2	3	7	4	6	2	9
6	6	9	8	7	4	3
3	4	8	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3 ³	4 ⁴	6 ⁴
3	2	4	1	9 ¹	8 ⁰	3 ²
0	1	3	9	2 ⁻¹	1 ⁰	4 ³

7x7

3	4	4
1	0	2
-1	0	3

3x3

Stride = 2

=

91	100	83
69	91	127
44	72	74

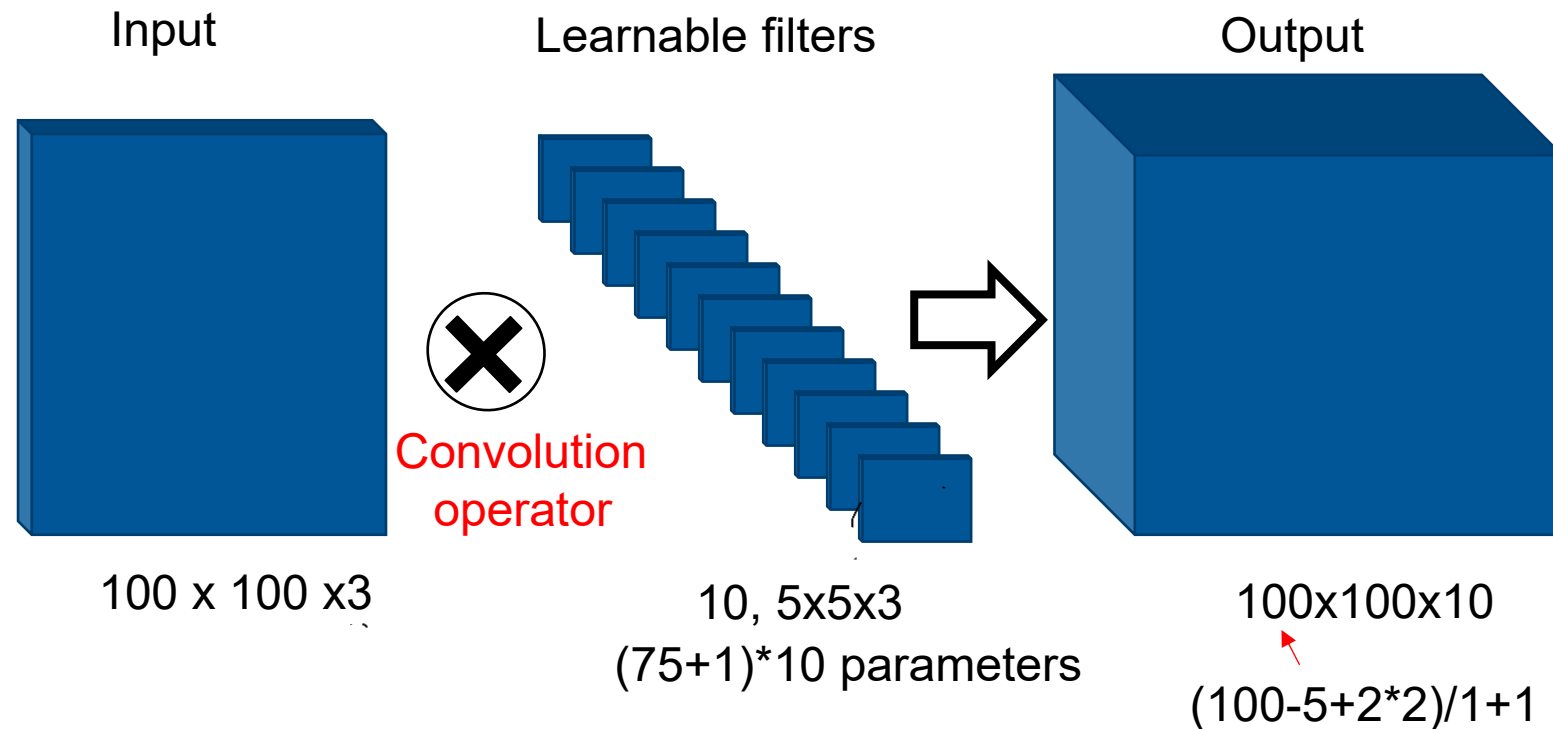
3x3

$n \times n$ * $f \times f$
 padding p stride s
 $s = 2$

$$\frac{n + 2p - f}{s} + 1 \quad \times \quad \frac{n + 2p - f}{s} + 1$$

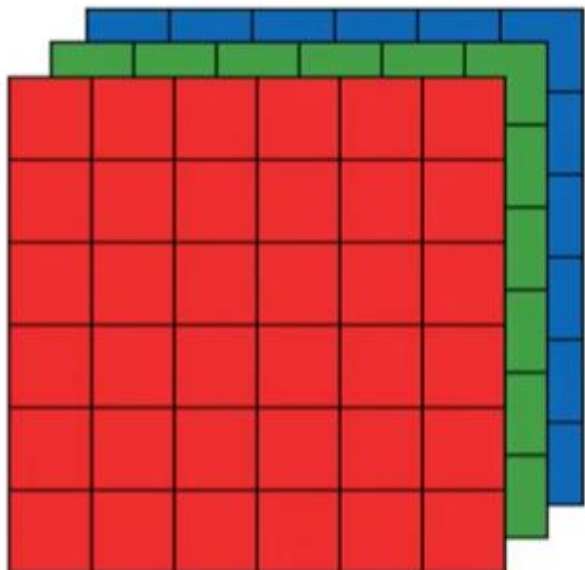
Convolution Operator

- Convolve the input image with a set of learnable small size filters
- Input size(N); filter size (F); zero padding (P); stride (S);
- Parameters to be optimised; output feature map size calculation $(N-F+2P)/S+1$;



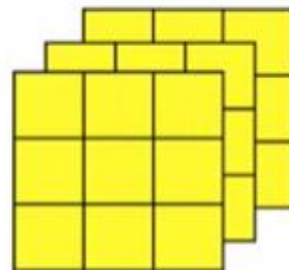


Convolution on RGB Images



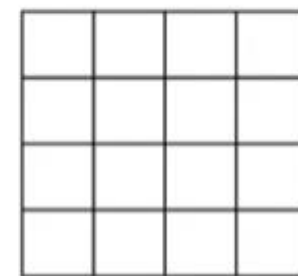
$6 \times 6 \times 3$

*



$3 \times 3 \times 3$

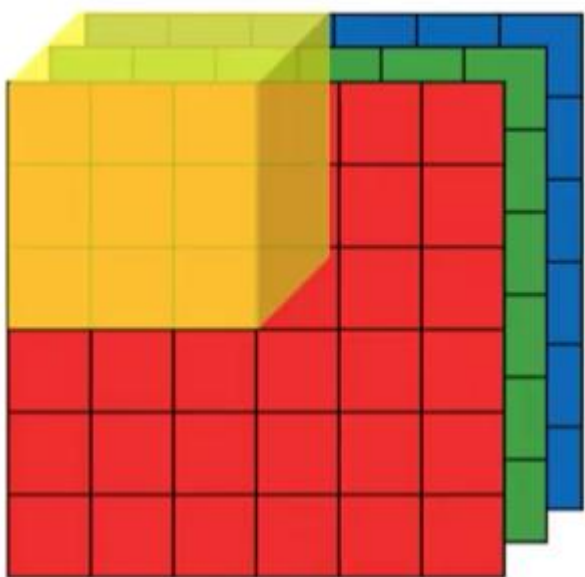
=



4×4

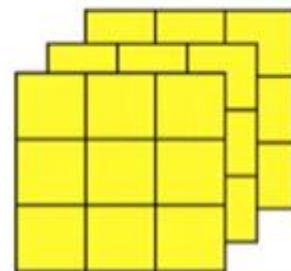


Convolution on RGB Images



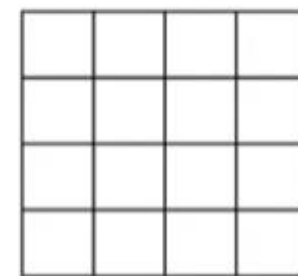
$6 \times 6 \times 3$

*



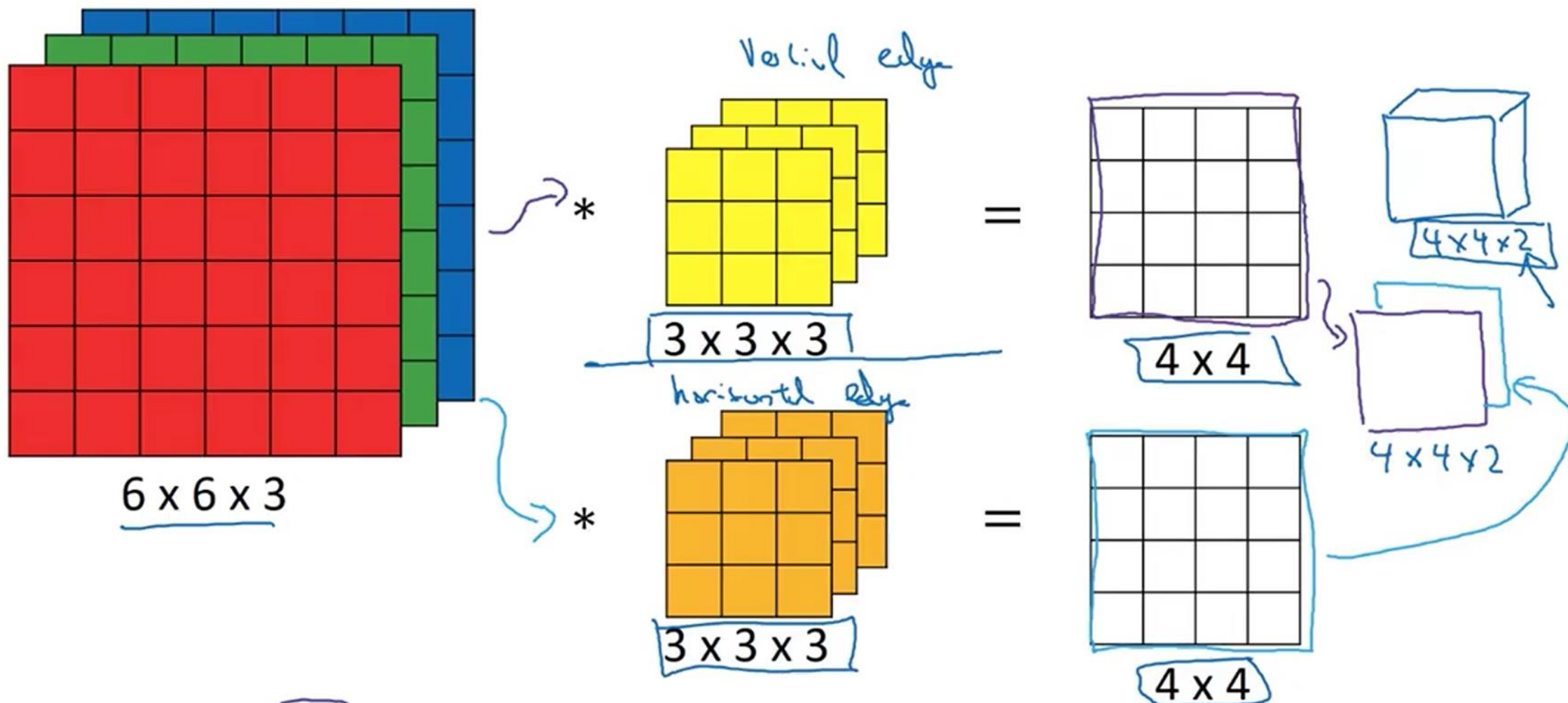
$3 \times 3 \times 3$

=



4×4

Convolution on RGB Images



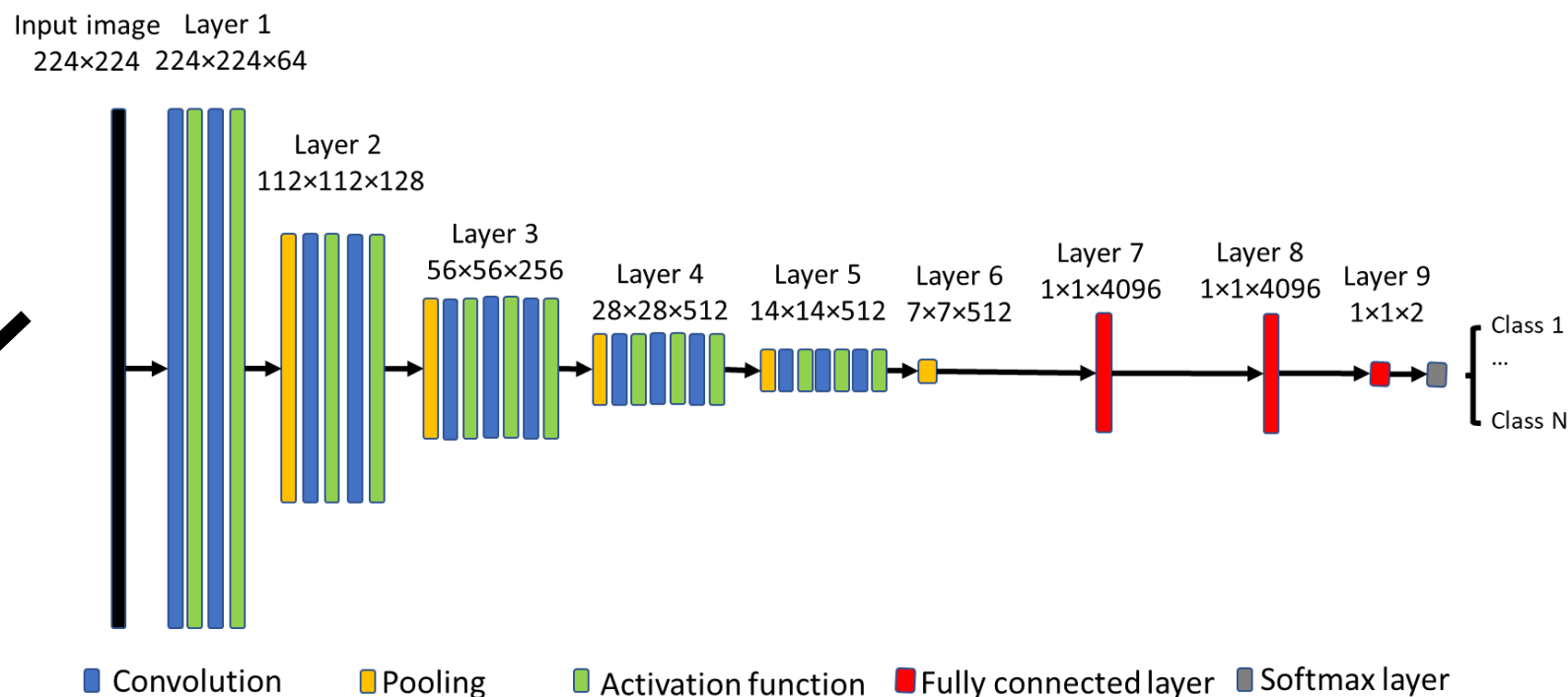
Summary: $n \times n \times n_c$ $\times$ $f \times f \times n_c$ $\rightarrow$ $n-f+1 \times n-f+1 \times n'_c$
 $6 \times 6 \times 3$ $3 \times 3 \times 3$ $4 \times 4 \times 2$ $\uparrow$ #filters



Basic Components of Convolutional Neural Network

- Convolution operator ✓
- Pooling
- Activation function ✓
- Fully connected layer ✓
- Loss functions ✓
- Optimisation methods

Example Deep CNN for classification





- What is a Convolutional Neural Network?
- What is a convolution?
- How do we handle color images with neural networks?
- What is vanishing gradient?



Next Lecture Learning Objectives

- Look into vanishing gradient and exploding gradient in more detail
- What is max-pooling?
- How many parameters in a CNN?
- Optimizing a CNN
- Gradient Descent recap
- Gradient Descent with Momentum
- RMSProp
- Adam